

# MingYu Zhao

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## EDUCATION

Rutgers University, Graduate School of Arts and Sciences

Ph.D. in Computer Science

University of Science and Technology of China

B.Eng. in Artificial Intelligence

New Brunswick, NJ

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Hefei, China

Jun 2024

## PUBLICATIONS “\*” denotes equal contribution.

### First-author Publications

- **Mingyu Zhao\***, Y. Zhou\*, Z. Wang, D. Gu, B. Guo, R. Ye, L. Han, C. Jin, D. N. Metaxas. “M3-Bench: Multi-Modal, Multi-Hop, Multi-Threaded Tool-Using MLLM Agent Benchmark.” Submitted to the *NeurIPS*, 2026.
- **Mingyu Zhao**, Z. Yang, Y. Zhou, Z. Xia, C. Jin, X. He, C. Neidle, D. N. Metaxas. “MHB: Multimodal Handshape-aware Boundary Detection for Continuous Sign Language Recognition.” Accepted by the *LREC*, 2026.
- **Mingyu Zhao\***, Z. Li\*, X. Dong, H. Ling, B. Huang. “CAPAA: Classifier-Agnostic Projector-Based Adversarial Attack.” Accepted by the *IEEE ICME*, 2025.

### Selected Other Publications

- “Lor-VP: Low-rank Visual Prompting for Efficient Vision Model Adaptation.” *ICLR* 2025.
- “Large Sign Language Models: Toward 3D American Sign Language Translation.” *WACV* 2026.
- “LED: LLM-Enhanced Open-Vocabulary Object Detection without Human-Curated Data Generation.” *CVPR* 2026.

## RESEARCH EXPERIENCE

**Topic: Multi-Hop, Multi-Threaded MLLM Agent Evaluation and Optimization** Oct. 2025 ~ Apr. 2026

**Instructor: Prof. [Dimitris N. Metaxas](#) (Rutgers University)**

- Built an end-to-end MCP benchmark generation and correctness validation framework; designed an MCP benchmark suite with 200+ tasks involving visual grounding and cross-tool dependencies, each with a standardized trajectory.
- Proposed a structure-aware tool-call evaluation metric using **sentence-encoder** embeddings and **similarity-bucketed Hungarian matching**; quantitatively evaluated GPT, Gemini, Claude, and 10+ MLLM models across the benchmark, with qualitative and quantitative analyses of budget and error types to characterize tool-call failure modes.
- Performed **supervised fine-tuning** on LLaDA 1.5 using generated tool-call data from multiple datasets, integrated a reasoning module, and optimized the model with **reinforcement learning**, improving specified tool-call task performance by **32%** over the base LLaDA 1.5 tool-calling baseline under **LLM judge** scores and accuracy metrics.

**Topic: Multi-Modal Video Temporal Boundary Detection**

Jan. 2025 ~ Dec. 2025

**Instructor: Prof. [Dimitris N. Metaxas](#) (Rutgers University) and Prof. [Carol Neidle](#) (Boston University)**

- Designed a spatio-temporal convolution-based segmentation module with velocity and acceleration features from skeleton sequences, improving **temporal boundary detection** F1 score by around **10%** over previous methods.
- Integrated handshape information by pretraining a handshape classifier over canonical categories and fusing the classifier’s learned hidden representations into the segmentation stream via **cross-attention** with a gating mechanism.
- Built an **end-to-end** continuous sign language recognition pipeline by combining boundary detection with sign recognition, improving boundary detection F1 score by nearly **20%** overall; for signs with well-localized boundaries, continuous recognition performance was within only **10%** of isolated-sign recognition.

**Topic: Classifier-Agnostic Projector-Based Adversarial Attack**

July. 2024 ~ Dec. 2024

**Instructor: Prof. [HaiBin Ling](#) (Stony Brook University) and Prof. [BingYao Huang](#) (Southwest University)**

- Proposed a pose-agnostic projector-based attack algorithm using Grad-CAM++ to focus perturbations on salient physical regions, mitigating the effects of camera-pose changes and improving visual stealthiness by around **20%**.
- Designed a novel optimal loss function to generate projector-based adversarial attacks across multiple classifiers with **negligible performance degradation** compared with classifier-specific attacks.
- Validated the effectiveness of the proposed CAPAA by building a camera-projector system in the real world and by evaluating the attack success rate and stealthiness on 10 sets of ablation experiments.

## INDUSTRY PROJECT

**Topic: ASL Pal**

Sep. 2025 ~ Present

**Leads: [Beryl Breindel](#) (DawnSignPress) and Prof. [Dimitris N. Metaxas](#) (Rutgers University)**

- Fine-tuned a state-of-the-art isolated American Sign Language (ASL) recognition model on the company’s proprietary ASL dataset, achieving **85%** recognition accuracy and outperforming the original model.
- Engineered the backend inference pipeline with model compression, parallel execution, model preloading, and other techniques, reducing model runtime from **5 min** to **15 sec**.
- Integrated the models into the web platform with the engineering team and validated the system through user testing.

## TECHNICAL STRENGTHS

**Skills:** Python, PyTorch, Bash, C/C++, AI-assisted IDE, Agentic Coding, Linux, MCP, Git, Flask, Slurm, L<sup>A</sup>T<sub>E</sub>X.